AYUSHMAN SRIVASTAV

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PROFESSIONAL SUMMARY

Aeronautical Engineering student (B.Tech, 2023–27) at FGIET Rae Bareli with a CGPA of 7.7, passionate about space technology, satellite design, and aerospace simulation. Experienced in NASA GMAT satellite orbit simulation, CubeSat structural design, ANSYS FEA (structural & thermal), and IoT embedded systems. ISRO-certified in Remote Sensing and Digital Image Analysis. Actively seeking internships in aerospace design, space technology, and UAV/drone systems.

CORE COMPETENCIES

CAD & Simulation	Fusion 360, SolidWorks, AutoCAD, Onshape, ANSYS Workbench (Structural & Thermal), MATLAB Simulink, NASA GMAT
Programming	MATLAB, Python, C, C++, HTML/CSS
IoT & Embedded	ESP8266, Arduino IDE, Autodesk Tinkercad
Aerospace	CubeSat Design, FEA, CFD, Orbital Mechanics, Remote Sensing, Satellite Communications
Tools & Platforms	ANSYS 2025 R2, Fusion 360, Figma, OBS Studio, ThingSpeak, Telegram Bot API
Soft Skills	Technical Research, Documentation, Project Management, Solo & Team Project Delivery

WORK EXPERIENCE

Engineering Intern *Aug 2025 – Sep 2025*

Kalam Pragati — MakeX Program (ERA Foundation × AKTU) | *Lucknow, Uttar Pradesh*

- Participated in the Kalam Pragati MakeX internship — a competitive engineering innovation program powered by ERA Foundation in association with AKTU, Lucknow
- Worked on practical engineering challenges applying aeronautical concepts to real-world problem statements

Space Technology Intern *Aug 2024 – Sep 2024*

Space Technology and Aeronautical Rocketry (STAR) | *Surat, Gujarat*

- Developed and simulated satellite orbit trajectories using NASA's General Mission Analysis Tool (GMAT), enhancing mission planning accuracy for LEO missions
- Configured spacecraft parameters, force models, and propagators to optimize Low Earth Orbit (LEO) trajectories for mission-specific requirements
- Executed detailed simulations to analyze orbital behavior, ground tracks, and mission performance metrics
- Gained hands-on exposure to real-world space mission workflows and satellite systems engineering

PROJECTS

CubeSat Structure Design & Thermal Analysis — LEO Mission *Jan 2026 – Present*

6th Semester Project | Solo | *FGIET Rae Bareli*

- Designed a complete 1U CubeSat (10×10×10 cm) mechanical assembly in Fusion 360 — corner columns, base plate, PCB holder, lattice-cutout side walls, and upper lid per CubeSat Design Specification (CDS Rev. 14)
- Performed Static Structural FEA in ANSYS Workbench 2025 R2 under 10g launch load — Max Von Mises Stress: 6.55×10^{-4} MPa (safe limit: 276 MPa) | Max Deformation: 1.63×10^{-6} mm

- Conducted Steady-State Thermal Analysis — Hot case: +120.39°C to -20.13°C gradient | Cold case (eclipse): uniform -100°C
- Tools: Fusion 360, ANSYS Workbench 2025 R2, MATLAB (orbital simulation — in progress)

CardioGuard — Real-Time Heart Monitoring & Alert System *Aug 2025 – Sep 2025*

IoT Project | Kalam Pragati Internship | *Lucknow*

- Designed a wearable IoT health device using AD8232 ECG sensor and MAX30102 pulse oximeter for continuous real-time cardiac monitoring
- Integrated ThingSpeak IoT dashboard for real-time data visualization and a Telegram bot for automated emergency pre-hospital alerts during critical 'golden minutes'
- Led hardware integration including 3D-printed casing designed in Onshape — successfully built and tested a working prototype
- Tools: Arduino IDE, ESP8266, Onshape, ThingSpeak API, Telegram Bot API

EDUCATION

B.Tech — Aeronautical Engineering *Sep 2023 – Jul 2027*

Feroze Gandhi Institute of Engineering and Technology (FGIET), Rae Bareli | CGPA: 7.7

Intermediate (Class XII) *Apr 2021 – Feb 2023*

Kamla Nehru Institute of Child Education

Matriculation (Class X) *2017 – 2020*

Central Academy

CERTIFICATIONS

Drone Systems & Controls	NPTEL	*Nov 2025*
AutoCAD Certified	Autodesk / Udemy	*Aug 2025*
Space Tech for Disaster Risk Reduction	ISRO	*Jun 2025*
Remote Sensing & Digital Image Analysis	ISRO — IIRS	*Mar 2025*
MATLAB Certified	Great Learning	*Aug 2024*

TECHNICAL SKILLS

CAD/Design: Fusion 360 • SolidWorks • AutoCAD • Onshape • Autodesk Tinkercad **Simulation:** ANSYS Workbench • MATLAB • NASA GMAT

Programming: MATLAB • Python • C • C++ • HTML/CSS **IoT/Embedded:** Arduino • ESP8266 • ThingSpeak • Telegram API

LANGUAGES

Hindi: Native / Bilingual **English:** Professional Working Proficiency